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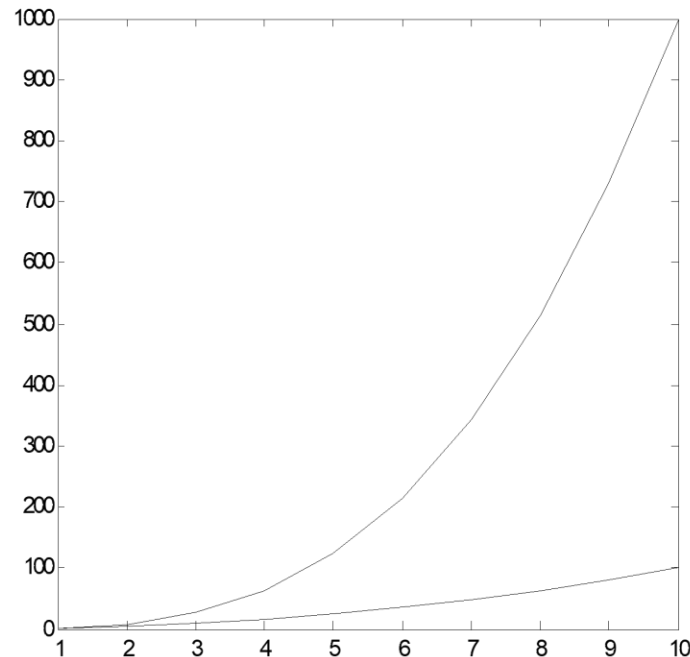
Module 6

Overview of Two-Dimensional Plotting – Part 2

Plot multiple curves on a figure

- Multiple plots can be added to a single figure by simply adding the x and y variables as additional arguments
- You can put as many additional plots as you want

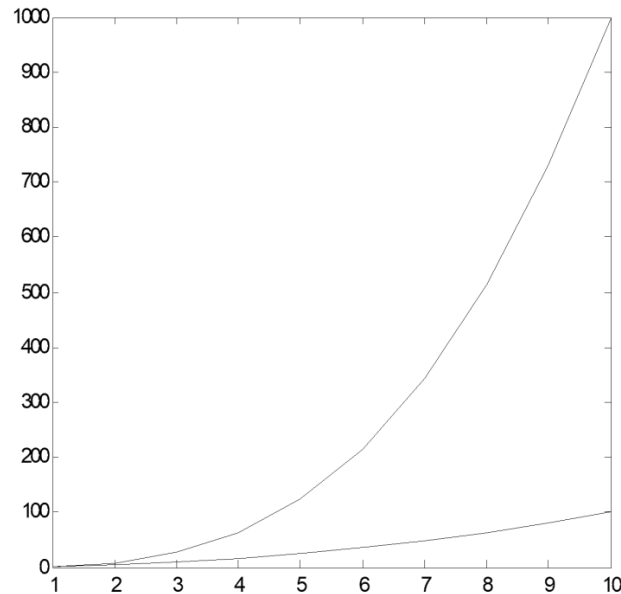
```
x = 1:1:10;  
y1 = x.^2;  
y2 = x.^3;  
plot(x,y1,x,y2);
```



The hold command

- Multiple plots can be added to a figure using the **hold** command
- The hold command follows the first plot function that you want to remain on the figure

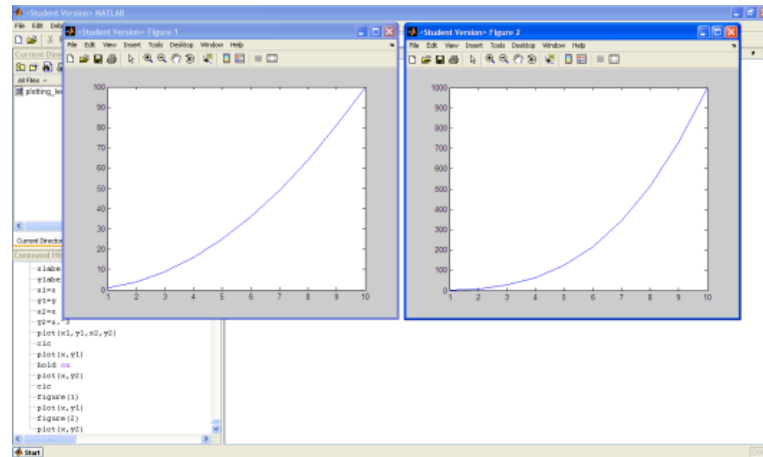
```
plot(x,y1);  
hold on;  
plot(x,y2);  
...  
hold off;
```



Multiple figures

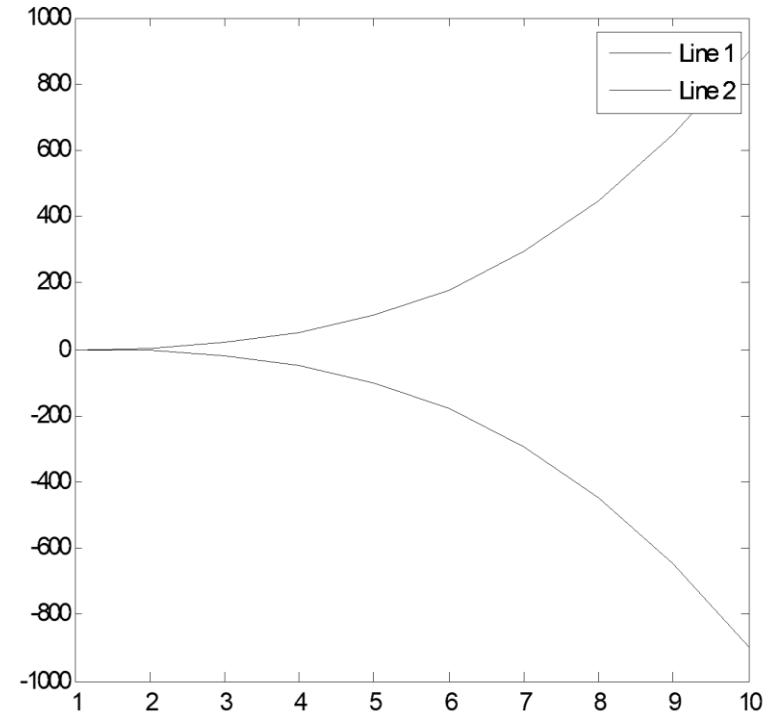
- Single (or multiple) plots can be created in multiple figures.
- The figure window can be specified using the figure command
 - e.g. selected figure window two requires figure(2) to be added before a plot function

```
figure(1);  
plot(x,y1);  
figure(2);  
plot(x,y2);
```



Addition of legends

```
x = 1:1:10;  
y1 = x.^3 - x.^2;  
y2 = (-1*x).^3 + x.^2;  
plot(x,y1,x,y2);  
legend('Line 1', 'Line  
2');
```



Special characters in legends

```
theta = -pi:0.01:pi;  
y = sin(theta);  
plot(theta,y);  
legend('y = sin(\theta)');
```

- Use the help feature to search for how to input other special characters.
- Search for 'text properties.'

